

Practice Test 2 (Challenge) · Answer Key

Physics Midterm · 80 points

How to use this key. This test is marked strictly on detail. A number without its unit, a vector without its direction or sign, or a diagram with one missing or extra force loses that point. In explanations, the underlined key terms earn the points.

Part A · Multiple Choice (2 points each, 36 total)

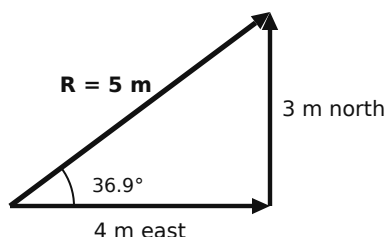
Quick key: 1 C 2 A 3 B 4 D 5 A 6 D 7 B 8 C 9 A 10 D 11 B 12 A 13 D 14 C 15 B
16 B 17 D 18 A

#	Ans	Why
1	C	speed. Speed has size only. Weight is a force, so it is a vector.
2	A	5.0 m. $\sqrt{(3.0^2 + (-4.0)^2)} = 5.0$ m. A magnitude is never negative.
3	B	The forces on it are balanced. Constant velocity means zero net force. Forces can still act, as long as they cancel.
4	D	kg·m/s². $F = ma$, so $1 \text{ N} = 1 \text{ kg} \times 1 \text{ m/s}^2$.
5	A	60 km/h. distance $\div$ time = $150/2.5 = 60$ km/h.
6	D	is 9.81 m/s² downward at every point. Only gravity acts, so the acceleration is g downward the whole time, including at the top.
7	B	-5.0 m/s². $72 \text{ km/h} = 20 \text{ m/s}$ (divide by 3.6). $a = (0 - 20)/4.0 = -5.0 \text{ m/s}^2$. -18 comes from not converting the units.
8	C	-5.0 N. $10 \cos 120^\circ = -5.0 \text{ N}$. In the second quadrant the x-component is negative. 8.7 N is the y-component.
9	A	-4.6 m/s. $v = u + at = 15 + (-9.81)(2.0) = -4.6 \text{ m/s}$. The negative sign means it is already moving down.
10	D	lands at the same time but twice as far out. Fall time depends only on the height: $t = \sqrt{(2h/g)}$. With the same t, $x = u_x t$ doubles.
11	B	3.0 m/s² east. $F_{\text{net}} = 10 - 4.0 = 6.0 \text{ N}$ east, so $a = 6.0/2.0 = 3.0 \text{ m/s}^2$ east. 7.0 comes from adding the forces.
12	A	589 N. Constant velocity means $a = 0$, so $N = mg = 60 \times 9.81 = 589 \text{ N}$. The 2.0 m/s is a velocity, not an acceleration.
13	D	17.3 m/s. At the top $v_y = 0$, but $v_x = 20 \cos 30^\circ = 17.3 \text{ m/s}$ is unchanged.
14	C	65°. Angles that add to 90° give the same range: $90^\circ - 25^\circ = 65^\circ$.
15	B	the two forces act on different objects; only the forces on the cart decide the cart's acceleration. Third-law forces never cancel because they act on different objects. The cart accelerates if the horse's pull on it is larger than the friction on it.
16	B	gravity only. The hand no longer touches the ball, so there is no "force of the throw". Inertia is not a force.
17	D	2.0 N. $a = 10/(4.0 + 1.0) = 2.0 \text{ m/s}^2$. The only horizontal force on B is from A: $F = (1.0)(2.0) = 2.0 \text{ N}$. 8.0 N is the net force on A.

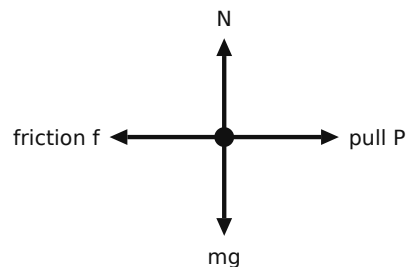
18	A	5.0 m at 90°. x: $5.0 + 0 - 5.0 = 0$. y: $0 + 5.0 + 0 = 5.0$ m. Resultant: 5.0 m at 90°.
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Part B · Free Response (44 points)

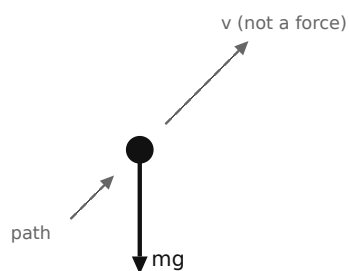
19. Find the mistakes (8 points)



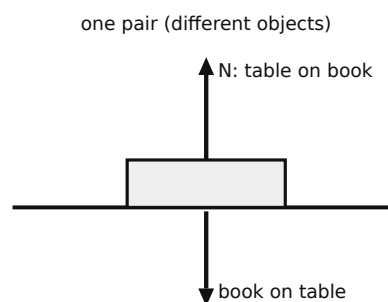
A. correct



B. correct



C. correct



D. correct

A. (1) The resultant points the wrong way: it must run from the **tail of the first** vector to the **tip of the last**. (2) The magnitude is not $4 + 3 = 7$ m: the vectors are perpendicular, so $R = \sqrt{4^2 + 3^2} = 5$ m, at 36.9° north of east.

B. (1) **Friction** is missing: it acts to the left. Because the velocity is constant, the net force is zero, so friction is equal in size to the pull. (2) The normal force arrow is shorter than the weight arrow: there is no vertical acceleration, so **$N = mg$** and the arrows must be the same length.

C. There is no "force of the throw" after the ball leaves the hand: a contact force needs **contact**. The only force is **gravity (mg) downward**. The ball keeps moving because of its inertia; velocity is not a force.

D. N and W both act on the **same object** (the book), and they are different types of force, so they are **not** a third-law pair. They are balanced forces. Correct pairs: table pushes up on book $\leftrightarrow$ book pushes down on table; Earth pulls down on book $\leftrightarrow$ book pulls up on Earth.

2 points each: 1 for stating the mistake(s), 1 for a correct diagram. For A and B both mistakes are needed for the first point.

20. Careful calculations (10 points)

(a) 1 point per row; both components need the correct sign and unit.

Vector	x-component	y-component	Quadrant check
25 m/s at 150°	$25 \cos 150^\circ = -21.7$ m/s	$25 \sin 150^\circ = +12.5$ m/s	II: x -, y +
40 N at 250°	$40 \cos 250^\circ = -13.7$ N	$40 \sin 250^\circ = -37.6$ N	III: x -, y -
12 m at 320°	$12 \cos 320^\circ = +9.19$ m	$12 \sin 320^\circ = -7.71$ m	IV: x +, y -
8.0 m/s ² at 90°	$8 \cos 90^\circ = 0$ m/s²	$8 \sin 90^\circ = +8.0$ m/s²	on the +y axis

(b) A: (6.0, 0) N. B: $(8.0 \cos 120^\circ, 8.0 \sin 120^\circ) = (-4.0, 6.93)$ N.

$R_x = 6.0 + (-4.0) = 2.0$ N, $R_y = 6.93$ N.

$R = \sqrt{(2.0^2 + 6.93^2)} = 7.21$ **N** at $\theta = \tan^{-1}(6.93/2.0) = 73.9^\circ$ from the +x axis.

1 point: B_x negative. 1 point: magnitude. 1 point: direction. Common slip: $B_x = +4.0$ gives $R = 12.2$ N, which earns at most 1 point.

(c) Up positive: $u = +12$ m/s, $a = -9.81$ m/s², $s = -20$ m (the ground is below the start).

$s = ut + \frac{1}{2}at^2$: $-20 = 12t - 4.905t^2$, so $4.905t^2 - 12t - 20 = 0$.

$t = [12 + \sqrt{(12^2 + 4(4.905)(20))}] / (2 \times 4.905) = (12 + 23.16)/9.81 = 3.58$ **s** (the negative root is rejected).

1 point: $s = -20$ m with $a = -9.81$ m/s². 1 point: correct quadratic. 1 point: positive root with unit. Splitting the motion into up and down parts is also accepted.

21. Projectile launched at an angle (10 points)

(a) $u_x = 25 \cos 40^\circ = 19.2$ **m/s** · $u_y = 25 \sin 40^\circ = 16.1$ **m/s**

(b) Back at $y = 0$: $T = 2u_y/g = 2(16.07)/9.81 = 3.28$ **s**

(c) At the top $v_y = 0$: $H = u_y^2/(2g) = 16.07^2/(2 \times 9.81) = 13.2$ **m**

(d) $R = u_x T = 19.15 \times 3.276 = 62.7$ **m** (check: $u^2 \sin 80^\circ/g = 62.7$ m)

(e) $v_x = 19.2$ m/s (unchanged). $v_y = u_y - gt = 16.07 - 9.81(1.0) = 6.26$ m/s.

$v = \sqrt{(19.15^2 + 6.26^2)} = 20.1$ **m/s** at $\tan^{-1}(6.26/19.15) = 18.1^\circ$ **above the horizontal** (still rising).

2 points per part: 1 for the method, 1 for the answer with unit. In (e): 1 point magnitude, 1 point direction. Keep unrounded values in the calculator; answers within ± 0.1 of these are accepted.

22. Explain using the correct key terms (8 points)

(a) The passengers have **inertia**. By **Newton's first law** they keep moving forward at the original velocity, because no net force acts on their upper bodies, while the bus slows down under them.

1 point: inertia / first law. 1 point: they continue at constant velocity while the bus slows. No credit for "a force throws them forward".

(b) Acceleration is the **rate of change of velocity**, not the velocity. **Gravity** still acts at the top, so the net force is mg and the acceleration is **9.81 m/s^2 downward**. The velocity is changing from upward to downward at that instant.

1 point: gravity (net force) still acts. 1 point: $a = 9.81 \text{ m/s}^2$ downward.

(c) By **Newton's third law** the forces on the truck and on the car are **equal in size** and opposite in direction. By **Newton's second law**, $a = F/m$, so the car, with its much **smaller mass**, has a much larger acceleration (change in velocity), and so more damage.

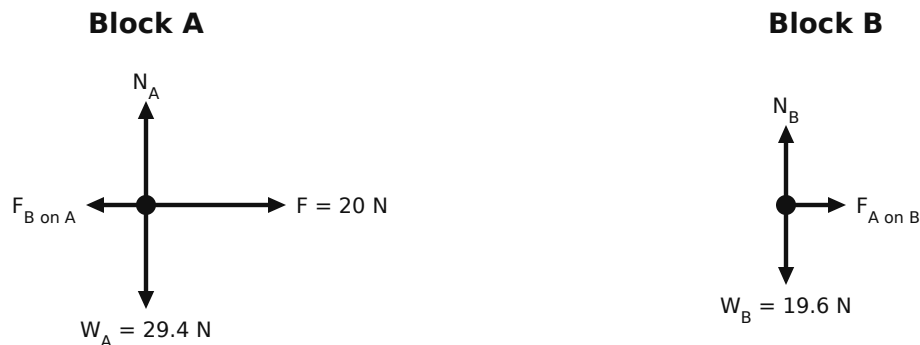
1 point: equal forces (third law). 1 point: smaller mass gives larger acceleration (second law).

(d) Your foot pushes **backward on the ground** (friction). By the third law the **ground pushes forward on your foot** with an equal force. That forward force acts **on you**, so it accelerates you forward.

1 point: the forward force is from the ground on you. 1 point: pair stated with both directions and both objects.

23. Newton's second and third laws together (8 points)

(a)



1 point: weight and normal on each block, equal lengths. 1 point: block A has $F = 20 \text{ N}$ to the right and the contact force from B to the left. 1 point: block B has only the contact force from A horizontally (the 20 N force does not act on B). The two contact-force arrows must be the same length.

(b) Treat both blocks as one system: $a = F/(m_A + m_B) = 20/(3.0 + 2.0) = \mathbf{4.0 \text{ m/s}^2}$ to the right

(c) For B alone: $F_{A \text{ on } B} = m_B a = (2.0)(4.0) = \mathbf{8.0 \text{ N to the right}}$. By the third law, $F_{B \text{ on } A} = \mathbf{8.0 \text{ N to the left}}$.

Check with A: $20 - 8.0 = 12 \text{ N} = (3.0)(4.0)$. ✓

(d) No. The acceleration is still 4.0 m/s^2 , but now the contact force is the only horizontal force on A: $F = m_A a = (3.0)(4.0) = \mathbf{12 \text{ N}}$. The contact force has to accelerate the block that the 20 N force does not touch.